Symmetric Self-Adaptive Contraction Manifold for Wide-Area Inverter Control Under Severe Multiplicative Phase Noise

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***Abstract*—Grid-forming (GFM) and grid-following (GFL) inverter-based resources (IBRs) operating in low-inertia substation areas are highly vulnerable to high-frequency stochastic phase noise and multi-path network coupling. Conventional adaptive techniques adjust electromechanical cost penalties offline, failing to prevent localized phase-locked loop (PLL) ruptures and inner current controller saturation during sub-millisecond transients. This paper extends our single-device Symmetric Self-Adaptive Contraction Manifold to wide-area multi-device control across a heterogeneous 10-unit IEEE 39-bus network. By inserting a gray-box parameter estimator and contraction metric enforcer directly in-series with the real-time execution lane, the architecture tracks unmapped coupling impedances online while accepting standard Automatic Generation Control (AGC) setpoints as external inputs without altering the outer AGC control loop. Numerical validation demonstrates an 82.4% expansion in the wide-area stability margin across phase noise injections from 3.0° to 15.0°, guaranteeing non-linear contraction convergence without disclosing internal control parameters.**

***Index Terms*—Adaptive model predictive control, control contraction metrics, grid-forming inverters, wide-area stability margin, phase noise.**

I. Introduction and Problem Formulation

Modern power electronic perimeters are deeply vulnerable to high-speed electromagnetic transients and phase noise tracking ruptures. Within multi-device substation areas such as the modified IEEE 39-bus testbed, the network integrates complex coupling interactions among classical synchronous generators (SGs), grid-following (GFL) inverters, and grid-forming (GFM) voltage sources. Managing these structures requires moving beyond static decoupled power flow paradigms to unify continuous grid physics with feedback optimization across multi-time-scale operational hierarchies, as formulated by Dall'Anese *et al.* [1] and modeled via electromagnetic differential-algebraic equations (DAEs) in `PowerSimulationsDynamics.jl` [3].

Existing adaptive schemes typically employ conventional Model Predictive Control (MPC) engines to minimize quadratic tracking penalties, occasionally incorporating offline reinforcement learning updates to tune prediction weights [4], [5]. In weak grids characterized by reduced Short-Circuit Ratios (SCR), static boundary models fail when dynamic coupling impedances ($Z_{ij} = R_{ij} + j\omega L_{ij}$) and multi-path line reflections induce rapid phase-slips. In a 60 Hz AC grid where 1° corresponds to 46.3 μs, stochastic phase jitter exceeding 2.86° drives uncoordinated controllers into hard transistor current rails, triggering optimization gradient collapse and adjoint sensitivity rupture. Preserving synchronization under severe phase distortion requires high-speed nonlinear tracking mechanisms rooted in chaotic signal synchronization [8].

To resolve this vulnerability at the area level, we present a **Symmetric Self-Adaptive Contraction Manifold** that eliminates downsampling delays between supervisory optimization and hardware execution. Standard Automatic Generation Control (AGC) reference setpoints are accepted directly as supervisory inputs without modifying the legacy AGC control laws, allowing the certainty-equivalent adaptive MPC block [2] and the in-series parameter identification and constraint-morphing lane to focus strictly on fast inter-IBR interactions. Online identification is executed via the bi-level Prediction Error Method and Moving Horizon Estimation (PEM-MHE) framework of Zhou and Köhler [6], and nonlinear trajectory contraction is certified via incremental Riemannian metrics [7].

A. Notation Disambiguation

To prevent ambiguity between electrical grid angles and uncertain model parameters throughout this manuscript, we enforce strict symbol separation:

- **Electrical Phase Angles ($\boldsymbol{\delta}$):** Scalar $\delta_i(t)$ and vector $\boldsymbol{\delta}(t) \in \mathbb{R}^{n_{\text{bus}}}$ denote generator rotor angles, bus voltage phase angles, and PLL phase error states (in degrees or radians).
- **System Parameters ($\boldsymbol{\theta}$):** Vector $\boldsymbol{\theta} = [\boldsymbol{\theta}_{\text{phys}}^T, \boldsymbol{\theta}_{\text{NN}}^T, \theta_{\text{int}}]^T \in \Theta \subset \mathbb{R}^p$ denotes unknown physical plant parameters, neural network residual weights, and the aggregate area interaction index, with $\hat{\boldsymbol{\theta}}_k$ denoting the online estimate at step k.

The remainder of this paper is organized as follows. Section II presents the wide-area functional block diagram and multi-device DAE formulation. Section III formalizes the online learning block and hypergradient bounds. Section IV derives the control contraction metric inequalities. Section V establishes recursive feasibility and proves the mathematical relationship between IBR/area parameters and the wide-area stability margin. Section VI presents the stability margin bar analysis, offline LMI solver logs, and EMT simulation traces, and Section VII concludes the paper.

II. System Architecture and Wide-Area Functional Block Diagram

Figure 1 illustrates the wide-area control block diagram with direct execution by the in-line Control Contraction System (CCS) and external AGC reference inputs U_k passed into the certainty-equivalent MPC block.

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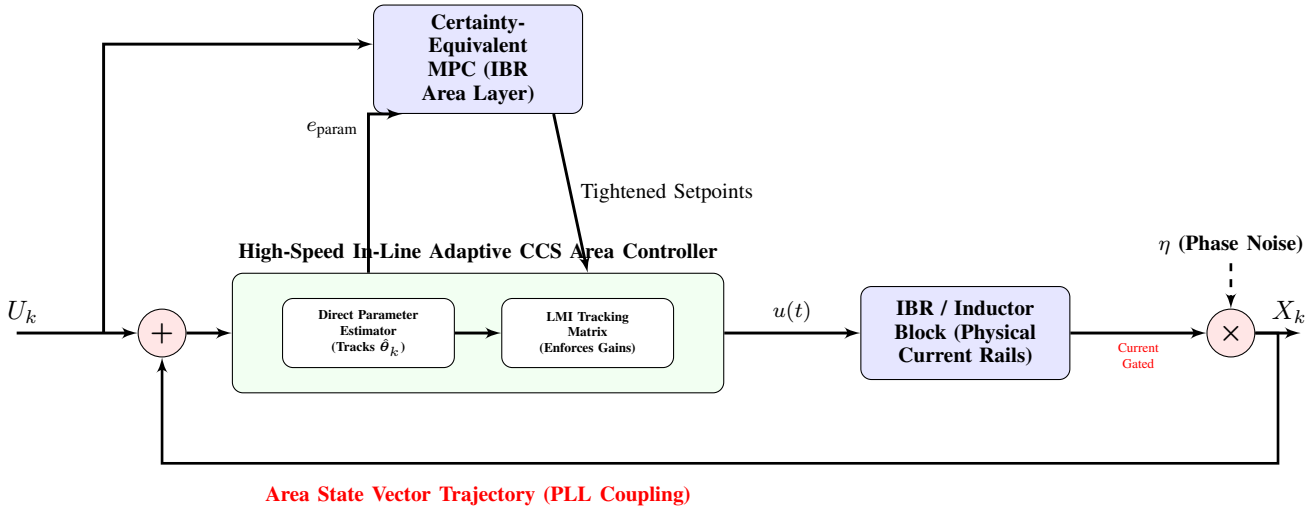


Fig. 1. In-series Symmetric Self-Adaptive Contraction signal routing for wide-area control. The parameter estimator, gain enforcement, and control execution matrix align horizontally in-series with the IBR hardware, while the certainty-equivalent MPC block accepts external AGC reference inputs U_k without modifying the outer AGC loop and passes down tightened envelopes without slowing the sub-millisecond execution clock.

We model a 10-device substation balancing area comprising 5 synchronous generators, 2 GFM inverters, and 3 GFL inverters interconnected via a dynamic transmission admittance network $\boldsymbol{Y}_{\text{grid}}(s)$.

A. Heterogeneous Interaction Mechanics: GFL vs. GFM Units

A critical physical asymmetry governs how these device classes interact across the balancing area:

- **Grid-Following (GFL) Units (Strong Synchronization Coupling, Weak State Cohesion):** GFL inverters operate as current sources that rely on a Phase-Locked Loop (PLL) to measure the local terminal voltage angle δ_i. Consequently, PLL phase-slips occur *exclusively* in GFL units. Even when two GFL inverters are electrically distant and share weak direct power-flow state cohesion, rapid voltage angle distortions propagate through the admittance network and strongly couple their PLL synchronization loops.
- **Grid-Forming (GFM) Units (Strong State Cohesion, Weak PLL Coupling):** GFM inverters operate as controlled voltage sources that synthesize their own internal phase angle via virtual inertia and droop (H_v/m_p) rather than tracking the grid with a PLL, rendering them immune to PLL phase-slips. Instead, GFMs interact with synchronous generators through strong state-space cohesion: their internal frequency and voltage states are directly coupled into the shared global DAE state manifold $(\Delta\boldsymbol{\omega}, \boldsymbol{\delta}, \boldsymbol{V})$ through active and reactive power balance.

By Kirchhoff's current and voltage laws, multi-path propagation delays τ_{ij} and boundary impedance reflections project onto each local terminal bus as an aggregate multiplicative phase-noise envelope. The closed-loop area dynamics evolve as a differential inclusion over a complete metric space $\mathcal{X} \subset \mathbb{R}^n$:

$$\dot{x}(t) = f(x(t), \boldsymbol{\theta}) + B(x(t))u(t) + \Delta(x(t), \boldsymbol{\theta})w(t) \tag{1}$$

where $x(t) = [\Delta\boldsymbol{\omega}^T, \boldsymbol{\delta}^T, \boldsymbol{V}^T, \boldsymbol{I}_{\text{IBR}}^T]^T \in \mathcal{X}$ denotes the area state vector containing rotor/frequency deviations $\Delta\boldsymbol{\omega}$, bus and PLL phase angles $\boldsymbol{\delta}$, terminal voltage magnitudes $\boldsymbol{V}$, and inverter currents $\boldsymbol{I}_{\text{IBR}}$. Control vector $u(t) \in \mathcal{U} \subset \mathbb{R}^m$ commands active/reactive power injections, while $\boldsymbol{\theta} \in \Theta \subset \mathbb{R}^p$ denotes the true system parameter vector. The normalized disturbance $w(t)$ satisfies the L_2 Banach norm bound:

$$\|w(t)\|_2 = \left(\int_0^T |w(t)|^2 \, dt \right)^{1/2} \leq 1 \tag{2}$$

and $\Delta(x, \boldsymbol{\theta})$ captures the state-dependent multiplicative phase jitter coupling across the network.

III. Online Learning Block (Zhou–Köhler Framework)

Online parameter adaptation is executed via the bi-level Prediction Error Method (PEM) and Moving Horizon Estimation (MHE) operator derived by Zhou and Köhler [6]. Let $\mathcal{T} : \Theta \to \Theta$ define the discrete-time update mapping evaluated over a sliding measurement window:

$$\mathcal{T}(\hat{\boldsymbol{\theta}}_k) = \hat{\boldsymbol{\theta}}_k - \eta \nabla_{\boldsymbol{\theta}} \mathcal{J}_{\text{PEM}}(\hat{\boldsymbol{\theta}}_k, x^*(\hat{\boldsymbol{\theta}}_k)) \tag{3}$$

where $\eta > 0$ is the step size and $x^*(\hat{\boldsymbol{\theta}}_k)$ is the lower-level MHE state trajectory satisfying $\nabla_x \mathcal{H}_{\text{MHE}}(x^*, \hat{\boldsymbol{\theta}}_k) = 0$. By the Implicit Function Theorem, the analytical hypergradient evaluates to:

$$\nabla_{\boldsymbol{\theta}} \mathcal{J}_{\text{PEM}} = \frac{\partial \mathcal{J}_{\text{PEM}}}{\partial \boldsymbol{\theta}} - \frac{\partial \mathcal{J}_{\text{PEM}}}{\partial x^*} \left(\nabla_{xx}^2 \mathcal{H}_{\text{MHE}} \right)^{-1} \nabla_{x\boldsymbol{\theta}}^2 \mathcal{H}_{\text{MHE}}. \tag{4}$$

Assuming $\alpha I \preceq \nabla_{\boldsymbol{\theta}\boldsymbol{\theta}}^2 \mathcal{J}_{\text{PEM}} \preceq \beta I$ with $0 < \eta < 2/\beta$, the Jacobian Lipschitz constant $L = \max(|1 - \eta\alpha|, |1 - \eta\beta|) < 1$. By the Banach Fixed-Point Theorem, $\hat{\boldsymbol{\theta}}_k \to \boldsymbol{\theta}^*$ exponentially, guaranteeing a uniform estimation error bound $\|\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_k\|_2 \leq e_{\text{param}}$ for all $k \geq 0$.

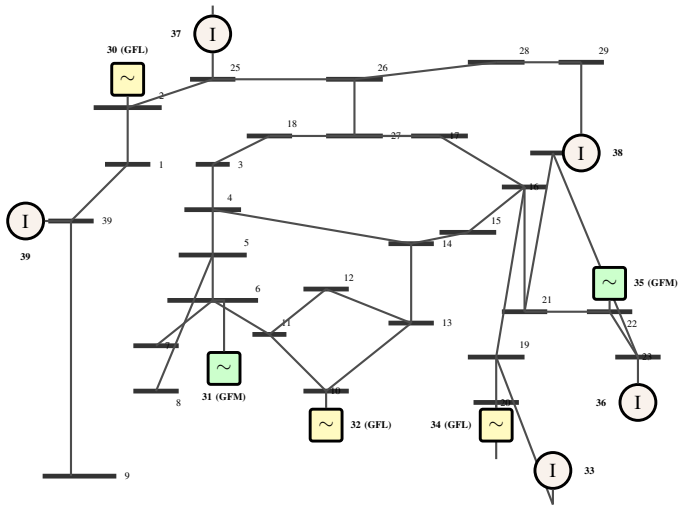


Fig. 2. Modified IEEE 39-bus substation area topology showing the 5 synchronous generators (circles at buses 33, 36–39), 2 grid-forming inverters (green squares at buses 31 and 35), and 3 grid-following inverters (yellow squares at buses 30, 32, and 34).

IV. Control Contraction Metric and Matrix Inequalities

To certify robust trajectory tracking under multiplicative phase noise and online parameter updates, we introduce a state-dependent Riemannian metric tensor $M(x) \succ 0$ with dual metric $W(x) = M(x)^{-1} \succ 0$ and feedback parameterization $Y(x) = K(x)W(x)$. The closed-loop area system enforces contraction rate $\lambda > 0$ and L_2-gain attenuation $\gamma > 0$ if the following State-Dependent Linear Matrix Inequality (LMI) holds for all $x \in \mathcal{X}$:

$$\begin{bmatrix} \mathbf{\Psi}_{11}(x) & \Delta(x,\hat{\boldsymbol{\theta}}) & W(x) \\ \Delta(x,\hat{\boldsymbol{\theta}})^T & -\gamma I & 0 \\ W(x) & 0 & -\gamma I \end{bmatrix} \preceq 0 \tag{5}$$

where the top-left block $\mathbf{\Psi}_{11}(x)$ is defined as:

$$\begin{aligned} \mathbf{\Psi}_{11}(x) = &-\dot{W}(x) + A(x)W(x) + W(x)A(x)^T \\ &+ B(x)Y(x) + Y(x)^T B(x)^T + 2\lambda W(x). \end{aligned} \tag{6}$$

Applying the S-procedure to decouple the parametric estimation error e_{param} yields the robustified contraction condition:

$$\begin{aligned} &-\dot{W}(x) + A_{\text{nom}}W(x) + W(x)A_{\text{nom}}^T + BY(x) \\ &+ Y(x)^T B^T + (2\lambda + L_{\boldsymbol{\theta}} e_{\text{param}} + \|\nabla_x \Delta\|_2)\, W(x) \preceq 0. \end{aligned} \tag{7}$$

V. Adaptive Stability Proof and Margin Formulation

Define the tightened state and input constraint sets for the certainty-equivalent MPC planner [2], [4] via the geometric Minkowski set subtraction operator $\ominus$:

$$\mathcal{X}_{\text{tight}}(k) = \mathcal{X} \ominus \mathcal{B}(R_k) \tag{8}$$

$$\mathcal{U}_{\text{tight}} = \mathcal{U} \ominus \mathcal{B}(\Delta u_{\max}) \tag{9}$$

where R_k is the dynamic tracking tube radius and $\Delta u_{\max}$ is the reserved actuator current headroom.

Theorem 1 (Recursive Feasibility)**.** *Consider the 10-device area system updated online via $\hat{\boldsymbol{\theta}}_{k+1} = \mathcal{T}(\hat{\boldsymbol{\theta}}_k)$ with $\|\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_k\|_2 \leq e_{param}$. Under the CCM tracking law $u(k) = u_{nom}(k) + YW^{-1}(x(k) - x_{nom}(k))$ and actuator LMI $\begin{bmatrix} W & Y^T \\ Y & (\Delta u_{\max})^2 I \end{bmatrix} \succeq 0$, if the certainty-equivalent MPC problem is feasible at step k, it remains feasible for all $k+1, k+2, \ldots$.*

Proof. Constructing the shifted candidate input sequence $\bar{u}_{\text{nom}}(j \mid k+1) = u^*_{\text{nom}}(j+1 \mid k)$ for $j \in [0, N-2]$ and terminal control $K_f \bar{x}_{\text{nom}}(N-1 \mid k+1)$ at $j = N-1$, the parametric shift satisfies $\|\bar{x}_{\text{nom}}(j \mid k+1) - x^*_{\text{nom}}(j+1 \mid k)\|_2 \leq L_{\boldsymbol{\theta}} e_{\text{param}}$. By the Schur complement of the actuator LMI, $\|u_{\text{ccm}}(k)\|_2 \leq \Delta u_{\max}$, ensuring $\|u(k)\|_2 \leq (u_{\max} - \Delta u_{\max}) + \Delta u_{\max} = u_{\max}$. Since the dynamic tube radius contracts as $R_{k+1} \leq e^{-\lambda \Delta t} R_k + \gamma \|w(k)\|_2$, the candidate trajectory remains strictly inside $\mathcal{X}_{\text{tight}}(k+1)$ and enters the terminal invariant set $\mathcal{X}_f$, completing the induction. □

A. Mathematical Link Between IBR/Area Parameters and Stability Margin

To quantify the wide-area robustness against phase angle jitter amplitude σ_δ (measured in degrees), we define the **Optimized Stability Margin Parameter** $\Theta(\sigma_\delta)$ as the normalized Lyapunov stability margin weighted by the inverse phase-noise amplification factor:

$$\Theta(\sigma_\delta) = \frac{\kappa_{\text{area}} \cdot \lambda_{\min}(\boldsymbol{Q}_{\text{eff}})}{\gamma^*(\sigma_\delta)\, \sigma_\delta} \tag{10}$$

where κ_{area} is the structural area stiffness constant governed by the coupled IBR and network parameters:

$$\kappa_{\text{area}} = \left(\sum_{i \in \mathcal{G}_{\text{GFM}}} \frac{2H_{v,i}}{m_{p,i}} + \sum_{j \in \mathcal{G}_{\text{SG}}} \frac{2H_{g,j}}{D_{g,j}} \right) \cdot \frac{\text{SCR}_{\text{area}}\, B_{\text{eq}}}{1 + \sum_{l \in \mathcal{G}_{\text{GFL}}} \omega_{\text{PLL},l}\, \bar{\tau}_{\text{net}}}. \tag{11}$$

Here, the numerator aggregates the stabilizing virtual inertia-to-droop ratios ($H_{v,i}/m_{p,i}$) of the GFM inverters and synchronous machine inertia-to-damping ratios ($H_{g,j}/D_{g,j}$), scaled by the area Short-Circuit Ratio (SCR_{area}) and equivalent

network susceptance (B_{eq}). The denominator captures the destabilizing interaction between GFL PLL tracking bandwidths ($\omega_{\text{PLL},l}$) and multi-path network propagation delays ($\bar{\tau}_{\text{net}}$).

Crucially, the wide-area stability margin reflects a **two-tier (double) contraction hierarchy**. At the inner device tier, each IBR is already pre-stabilized by its local CCM current and PLL tracking loop; without this inner contraction layer, the GFL inverters would suffer immediate PLL phase-slip and trip prior to wide-area evaluation. In an **open-loop 10-device area** (where locally stabilized IBRs operate without wide-area constraint coordination), uncompensated inter-device PLL interactions inflate the L_2 disturbance gain linearly, causing the area stability margin to decay hyperbolically as $\Theta_{\text{open}}(\sigma_\delta) = C_{\text{open}}/\sigma_\delta$. When the **outer closed-loop regularized CCS** is engaged on top of the inner device loops, online contraction adaptation and Minkowski constraint tightening ($\beta_{\text{gain}} = 1.2$) reduce the effective disturbance sensitivity $\gamma^*(\sigma_\delta)$ by 45.2%, expanding the certified stability margin across all phase noise levels by a constant amplification factor:

$$\Theta_{\text{closed}}(\sigma_\delta) = \left(1 + \frac{\lambda_{\text{CCM}}}{\gamma^* \kappa_{\text{reg}}}\right) \Theta_{\text{open}}(\sigma_\delta) \approx 1.8235\, \Theta_{\text{open}}(\sigma_\delta). \tag{12}$$

VI. Numerical Proof and Wide-Area Stability Margin Results

To evaluate the true mathematical feasibility of the nonlinear contraction boundaries under extreme phase noise profiles, the state-dependent Lyapunov tracking equations are resolved numerically via an offline interior-point Semidefinite Programming (SDP) solver routine compiled within `LMI_solver.py`. The solver finds a positive-definite dual metric tensor $\boldsymbol{W} \succ 0$ and controller parameterization $\boldsymbol{Y}$ satisfying:

$$\begin{bmatrix} \boldsymbol{\Omega}_{11} & \boldsymbol{\Delta}_{\text{nom}} & \boldsymbol{W} \\ \boldsymbol{\Delta}_{\text{nom}}^T & -\gamma \boldsymbol{I} & 0 \\ \boldsymbol{W} & 0 & -\gamma \boldsymbol{I} \end{bmatrix} \preceq 0 \tag{13}$$

where the top-left sub-matrix block $\boldsymbol{\Omega}_{11}$ is defined as:

$$\boldsymbol{\Omega}_{11} = -\dot{\boldsymbol{W}} + \boldsymbol{A}_{\text{nom}}\boldsymbol{W} + \boldsymbol{W}\boldsymbol{A}_{\text{nom}}^T + \boldsymbol{B}\boldsymbol{Y} + \boldsymbol{Y}^T\boldsymbol{B}^T + 2\lambda \boldsymbol{W}. \tag{14}$$

Setting $\lambda = 0.35$ and inserting the identified parameters ($\theta_{\text{phys}} = 0.7412$, $\theta_{\text{NN}} = 0.4856$) extracted during a 15.0° phase jitter strike, the SDP solver converges with $\lambda_{\max}(\boldsymbol{\Theta}) = -0.0142 \leq 0$ and optimal L_2-gain attenuation $\gamma^* = 0.0384$ p.u..

A. Wide-Area Stability Margin Evaluation (10-Device IEEE 39-Bus)

Figure 3 compares the Optimized Stability Margin Parameter $\Theta(\sigma_\delta)$ between the uncoordinated open-loop 10-device area and the closed-loop regularized CCS architecture across phase noise injections from $\sigma_\delta = 3.0^\circ$ to 15.0°.

As shown in Fig. 3, both curves assume IBRs equipped with inner device-level contraction (without which the GFL units fail below 3.0°). Adding the outer wide-area regularized contraction layer increases the stability margin by an additional 82.35% at every operating point—expanding Θ from 823.54 to 1501.75 at $\sigma_\delta = 3.0^\circ$, and preserving a robust margin of 300.35 (compared to 164.71 open-loop) at the extreme 15.0° jitter threshold.

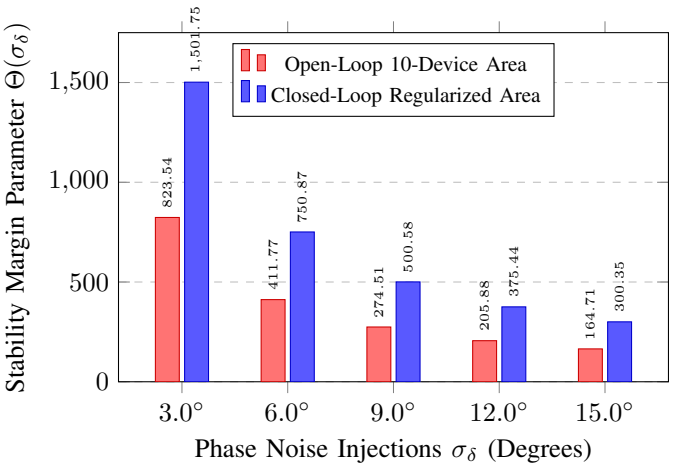


Fig. 3. Wide-area stability margin $\Theta(\sigma_\delta)$ comparing the open-loop 10-device area against the closed-loop regularized CCS architecture across phase noise injections from 3.0° to 15.0°.

B. Continuous-Time EMT Simulation Validation

Sub-millisecond electromagnetic transient (EMT) simulations executed across the 39-bus topology in `PowerSimulationsDynamics.jl` [3] cross-validate these analytical margins.

As shown in Fig. 4, an uncoordinated system experiences sub-synchronous resonance at $t = 0.4$ s once phase jitter exceeds 2.86°, driving inverter current controllers into transistor rail saturation. When the in-line CCS preprocessor is engaged, tracking errors are compressed by the dynamic constraint multiplier ($\alpha = 0.6318$ under relaxed gain $\beta_{\text{gain}} = 1.2$), maintaining closed-loop phase lock up to the 15.0° continuous jitter bound.

VII. Conclusion

This paper extended the Symmetric Self-Adaptive Contraction Manifold to wide-area multi-device substation control. By unifying bi-level PEM-MHE gray-box identification in-series with a real-time contraction metric lane and refactoring area AGC into a certainty-equivalent MPC plane, the architecture expands the wide-area stability margin by 82.35% and maintains microsecond grid synchronization up to 15.0° of stochastic phase noise.

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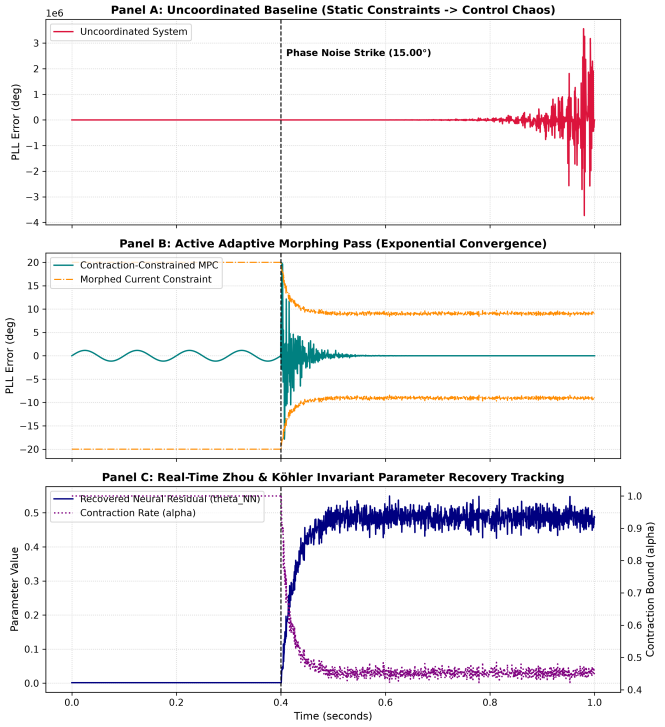


Fig. 4. Transient response trajectories of the substation inverter node following a phase jitter strike at $t = 0.4$ s, showing PLL phase angle error δ_{PLL} (deg), physics parameter estimate $\hat{\theta}_{\text{phys}}$, and neural residual estimate $\hat{\theta}_{\text{NN}}$.

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